

5.7 Grinding

5.7.1 Grinding action

Industrial grinding machines used in the mineral processing industries are mostly of the tumbling mill type. These mills exist in a variety of types – rod, ball, pebble autogenous and semi-autogenous. The grinding action is induced by relative motion between the particles of media – the rods, balls or pebbles. This motion can be characterized as collision with breakage induced primarily by impact or as rolling with breakage induced primarily by crushing and attrition. In autogenous grinding machines fracture of the media particles also occurs by both impact (self breakage) and attrition.

The relative motion of the media is determined by the tumbling action which in turn is quite strongly influenced by the liners and lifters that are always fixed inside the shell of the mill. Liners and lifters have two main purposes:

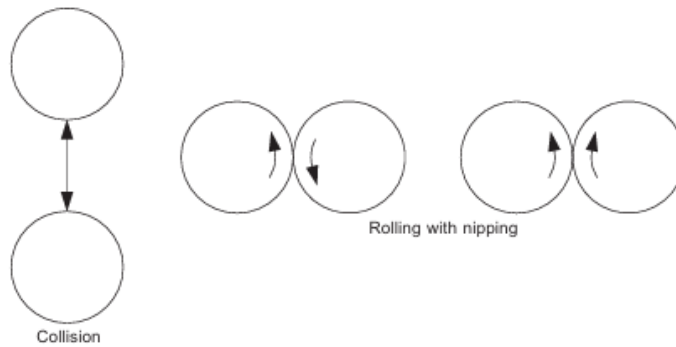


Figure 5.20 Different types of grinding action by the grinding media

1. Liners protect the outer shell of the mill from wear – liners are renewable.
2. Lifters prevent slipping between the medium and slurry charge in the mill and the mill shell. Slippage will consume energy wastefully but more importantly it will reduce the ability of the mill shell to transmit energy to the tumbling charge. This energy is required to cause grinding of the material in the mill. The shape and dimensions of the lifters control the tumbling action of the media.

The tumbling action is difficult to describe accurately but certain regions in the mill can be characterized in terms of the basic pattern of motion of material in the mill.

The motion of an individual ball in the charge is complicated in practice and it is not possible to calculate the path taken by a particular particle during rotation of the charge. However the general pattern of the motion of the media can be simulated by discrete element methods, which provide valuable information about the dynamic conditions inside the mill.

Some of the terms that are often used to describe the motion of the media in a tumbling mill are shown in Figure 5.21.

5.7.2 Critical speed of rotation

The force balance on a particle against the wall is given by

Centrifugal force outward

$$F_c = m_p \omega^2 \frac{D_m}{2} \quad (5.54)$$

ω is the angular velocity, m_p is the mass of any particle (media or charge) in the mill and D_m is the diameter of the mill inside the liners.

Gravitational force

$$F_g = m_p g \quad (5.55)$$

The particle will remain against the wall if these two forces are in balance, i.e.

$$F_c = F_g \cos \forall \quad (5.56)$$

where $\forall$ is shown in Figure 5.21.

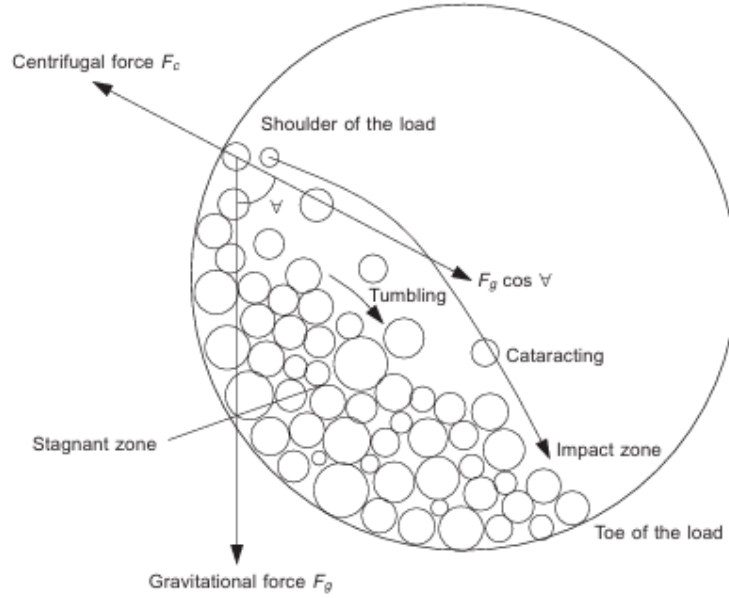


Figure 5.21 Media motion in the tumbling mill

Thus a particle will separate from the wall at the point where

$$\cos \forall = \frac{F_c}{F_g} \quad (5.57)$$

The critical speed of the mill, ω_c , is defined as the speed at which a single ball will just remain against the wall for a full cycle. At the top of the cycle $\forall = 0$ and

$$F_c = F_g \quad (5.58)$$

$$m_p \frac{\omega_c^2 D_m}{2} = m_p g \quad (5.59)$$

$$\omega_c = \left(\frac{2g}{D_m} \right)^{1/2} \quad (5.60)$$

The critical speed is usually expressed in terms of the number of revolutions per second

$$\begin{aligned}
 N_c &= \frac{\omega_c}{2\pi} = \frac{1}{2} \left(\frac{2g}{D_m} \right)^{1/2} = \frac{(2 \times 9.81)^{1/2}}{2\pi D_m^{1/2}} \\
 &= \frac{0.705}{D_m^{1/2}} \text{ revs/sec} \\
 &= \frac{42.3}{D_m^{1/2}} \text{ revs/min}
 \end{aligned} \tag{5.61}$$

The liner profile and the stickiness of the pulp in the mill can have a significant effect on the actual critical velocity. Mills usually operate in the range 65–82% of critical but values as high as 90% are sometimes used.

A parameter that defines the performance of a mill is the energy consumption. The power supplied to the mill is used primarily to lift the load (medium and charge). Additional power is required to keep the mill rotating.

5.7.3 Power drawn by ball, semi-autogenous and autogenous mills

A simplified picture of the mill load is shown in Figure 5.22 and this can be used to establish the essential features of a model for mill power.

The torque required to turn the mill is given by:

$$\text{Torque} = T = M_c g d_c + T_f \tag{5.62}$$

where M_c is the total mass of the charge in the mill and T_f is the torque required to overcome friction.

$$\text{Power} = 2\pi NT \tag{5.63}$$

For mills of different diameter but running at the same fraction of critical speed and having the same fractional filling

$$\begin{aligned}
 \text{Net power} &= 2\pi N M_c d_c \\
 &= \alpha N_c M_c d_c \\
 &= \alpha \frac{1}{D_m^{0.5}} L D_m^2 D_m \\
 &= \alpha L D_m^{2.5}
 \end{aligned} \tag{5.64}$$

The exponent 2.5 on D_m has been variously reported to have values as low as 2.3 and as high as 3.0.

The effect of varying mill speed on the power drawn by the mill is shown graphically in Figure 5.26. The speed of rotation of the mill influences the power draft through two effects: the value of N and the shift in the center of gravity with speed. The center of gravity first starts to shift to the left as the

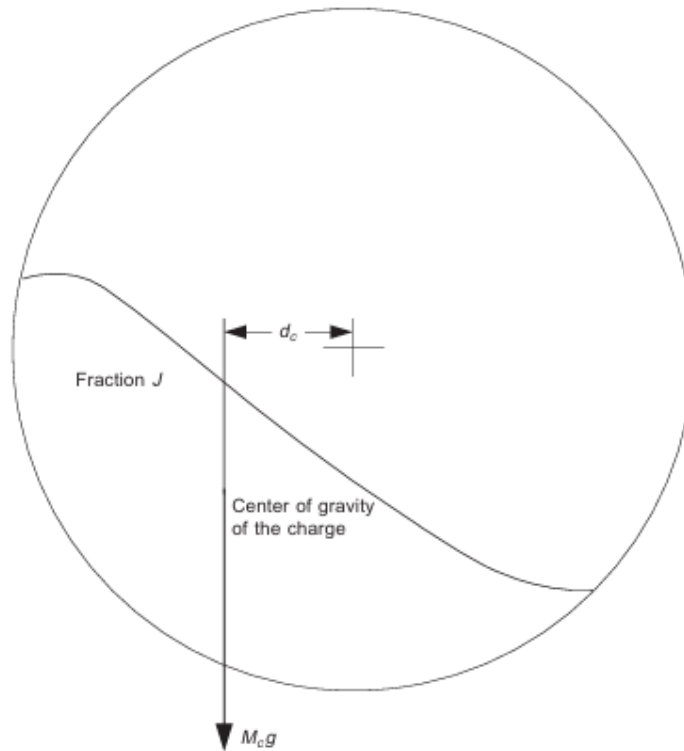


Figure 5.22 Simplified calculation of the torque required to turn a mill

speed of rotation increases but as critical speed is reached the center of gravity moves towards the center of the mill as more and more of the material is held against the shell throughout the cycle. Since critical speed is larger at smaller radii the centrifuging layer gets thicker and thicker as the speed increases until the entire charge is centrifuging and the net power draw is zero.

The effect of mill charge is primarily through the shifting of the center of gravity and the mass of the charge. As the charge increases the center of gravity moves inward. The power draft is more or less symmetrical about the 50% value.

A simple equation for calculating net power draft is:

$$P = 2.00 \varphi_c D_m^{2.5} L K_l \text{ kW} \quad (5.65)$$

K_l is the loading factor which can be obtained from Figure 5.23 for the popular mill types. φ_c is the mill speed measured as a fraction of the critical speed.

More reliable models for the prediction of the power drawn by ball, semi-autogenous and fully autogenous mills have been developed by Morrell (1996) and by Austin (1990). The relationship between gross power and net power is given by

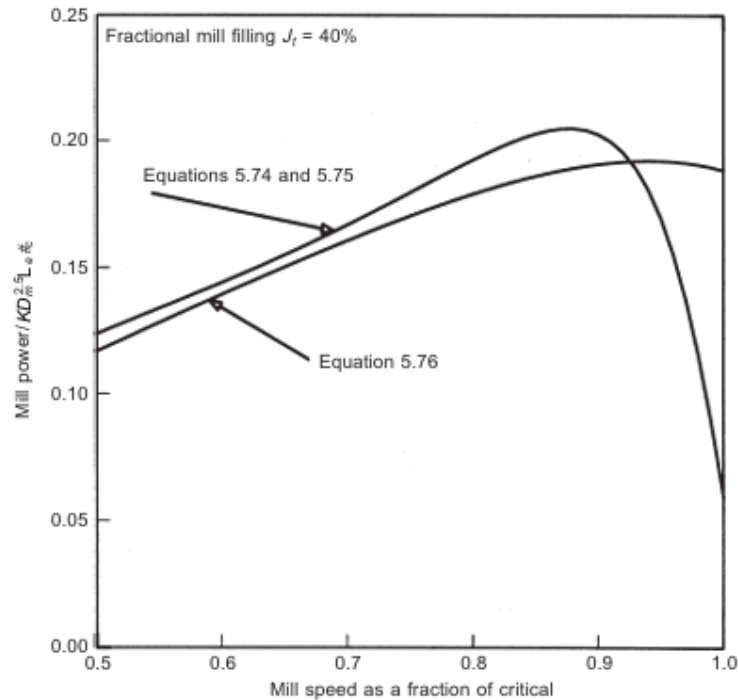


Figure 5.26 The effect of mill speed on the power drawn by a rotating mill

the motion of the media particles including all the many collisions in an operating mill. The distribution of impact energies is called the impact energy spectrum of the mill and this distribution function ultimately determines the kinetics of the comminution process in the mill.

5.7.6 Kinetics of breakage

The process engineering of milling circuits is intimately linked with the kinetic mechanisms that govern the rate at which material is broken in a comminution machine. The discussion of breakage phenomena in previous sections has been based on the concept of individual impact events in which a collection of parent particles are involved in a single rapid application of compressive stress in a mechanical event that has been loosely described as an impact. During each impact event each parent particle captures some of the available impact energy and as a result it may break. If it does break its immediate progeny will also absorb some of the residual impact energy and these progeny particles may or may not break. Any progeny particle that breaks will produce progeny that in turn may break. This repeated fracture of successive generations of progeny continues until all the energy of the original impact has been dissipated. Each parent has a unique set of progeny particles that result from

the impact and each progeny comes from one and only one original parent. The distribution of sizes of the set of progeny particles is defined by the breakage function of the parent. These discrete impact events happen with high frequency in a grinding mill and it is convenient to consider the breakage as a continuous process. This provides an effective modeling method that is based on the population balance approach.

In any region of a comminution machine, the rate at which material of a particular size is being broken is a strong function of the amount of that size of material present. This is the basis of the model. The rate of breakage varies with size. It is convenient to use discrete size classes that provide an adequate approximation for practical computation.

The fundamental postulate that supports the kinetic model is that the rate of breakage of material out of size class i is proportional to the amount of material of size i in the mill. Thus

$$\text{Rate of breakage} = k_i M m_i$$

k_i is called the specific rate of breakage. It is also called the selection function. Note that k_i is a function of the representative size in the class. It may also be a function of the composition of the parent particle.

Material breaking out of class i distributes itself over all other classes according to the breakage function which is specific to the comminution operation and the material being broken.

It is common practice to normalize the breakage function to the lower mesh size of the interval so that breakage means breakage out of a size class. If the breakage function is described by the standard form:

$$B(x; y) = K \left(\frac{x}{y} \right)^{n_1} + (1 - K) \left(\frac{x}{y} \right)^{n_2} \quad (5.78)$$

then

$$\begin{aligned} b_{ij} &= B(D_{i-1}; D_j) - B(D_i; D_j) \\ b_{jj} &= 0 \end{aligned} \quad (5.79)$$

The relationship between the cumulative breakage function and b_{ij} is shown in Figure 5.27. The breakage function has the value 1.0 at the lower boundary of the parent size interval. This reflects the convention that breakage is assumed to occur only when material leaves the parent size interval. This convention has been adopted because it is only possible to detect breakage in a laboratory experiment if the material leaves the sieve interval. The convention is not an essential feature of the population balance method but it is convenient because most of the breakage data that appears in the literature has it as the base. At all events it is not difficult to convert the breakage function data to the alternative convention if required. The differences between Equation 5.79 and the equivalent expressions for crushing machines (5.52 and 5.53) should be noted.

If normalization is satisfactory and if a strictly geometric progression is

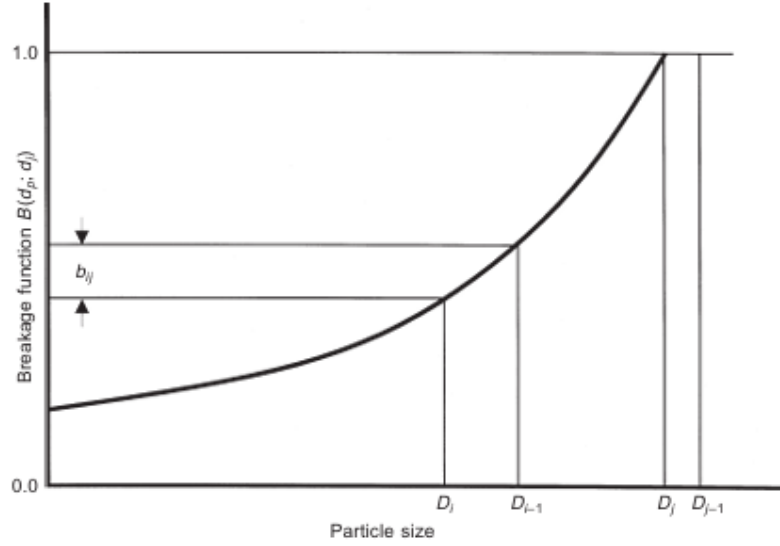


Figure 5.27 The breakage function used for the description of milling operations in rotating mills. Compare with the equivalent diagram for crushing machines (Figure 5.19)

used for the mesh size then b_{ij} is a function only of the difference $i - j$. For example if a root 2 series is used for the size classes.

$$\frac{D_i}{D_j} = \frac{D_i}{(\sqrt{2})^{i-j} D_i} = (\sqrt{2})^{j-i} \quad (5.80)$$

$$b_{ij} = K \left(\frac{D_{i-1}}{D_j} \right)^{n_1} + (1 - K) \left(\frac{D_{i-1}}{D_j} \right)^{n_2} - K \left(\frac{D_i}{D_j} \right)^{n_1} + (1 - K) \left(\frac{D_i}{D_j} \right)^{n_2} \quad (5.81)$$

$$\begin{aligned} b_{ij} &= K[(\sqrt{2})^{n_1(j-i+1)} - (\sqrt{2})^{n_1(j-i)}] + (1 - K)[(\sqrt{2})^{n_2(j-i+1)} - \sqrt{2}^{n_2(j-i)}] \\ &= K(\sqrt{2})^{n_1(j-i)}(\sqrt{2} - 1) + (1 - K)(\sqrt{2})^{n_2(j-i)}(\sqrt{2} - 1) \\ &= 0.414K(\sqrt{2})^{n_1(j-i)} + 0.414(1 - K)(\sqrt{2})^{n_2(j-i)} \end{aligned} \quad (5.82)$$

5.8 The continuous mill

Industrial grinding mills always process material continuously so that models must simulate continuous operation. Suitable models are developed in this section.

5.8.1 The population balance model for a perfectly mixed mill

The equations that describe the size reduction process in a perfectly mixed ball mill can be derived directly from the master population balance equation that was developed in Chapter 2. However, the equation is simple enough to derive directly from a simple mass balance for material in any specific size class.

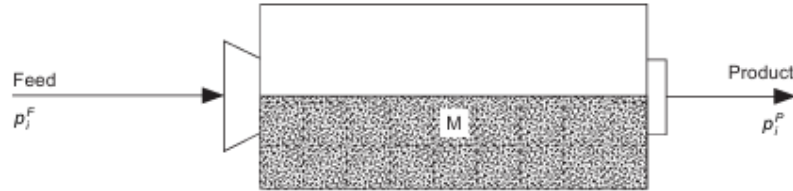


Figure 5.28 The perfectly mixed continuous mill

This simple equation is derived below using the following definitions:

- p_i^F = fraction feed in size class i
- p_i^P = fraction of product in size class i
- m_i = fraction of mill contents in size class i
- M = mass of material in the mill
- W = mass flowrate through the mill.

A mass balance on size class i is developed by noting that the contents of the mill receives material in size class i from the feed and from the breakage of material in the mill that is of size greater than size i . Material of size i is destroyed in the mill by fracture.

$$Wp_i^P = Wp_i^F + M \sum_{j=1}^{i-1} b_{ij}k_jm_j - Mk_im_i$$

$$p_i^P = p_i^F + \sum_{j=1}^{i-1} b_{ij}k_jm_j - k_i\tau m_i$$
(5.83)

where $\tau = M/W$ is the average residence time of the material in the mill.

If the contents of the mill are assumed to be perfectly mixed $m_i = p_i^P$

$$p_i^P = p_i^F + \tau \sum_{j=1}^{i-1} b_{ij}k_jp_j^P - k_i\tau p_i^P$$

$$p_i^P = \frac{p_i^F + \sum_{j=1}^{i-1} b_{ij}k_j\tau p_j^P}{1 + k_i\tau} \quad \text{for all } i$$
(5.84)

These can be solved by a straightforward recursion relationship starting with size class 1.

$$\begin{aligned}
 p_1^P &= \frac{p_1^F}{1 + k_1 \tau} \\
 p_2^P &= \frac{p_2^F + b_{21} k_1 p_1^P}{1 + k_2 \tau} \\
 p_3^P &= \frac{p_3^F + b_{31} k_1 p_1^P + b_{32} k_2 p_2^P}{1 + k_3 \tau}
 \end{aligned} \tag{5.85}$$

etc.

5.8.2 The mill with post classification

In practice the material in the mill does not have unrestricted ability to leave in the outlet stream. Larger particles are prevented from leaving by the discharge grate if one is present and even in overflow discharge mills, the larger particles do not readily move upward through the medium bed to the discharge. On the other hand very small particles move readily with the water and are discharged easily. Thus the discharge end of the mill behaves as a classifier which permits the selective discharge of smaller particles and recycles the larger particles back into the body of the mill. This is illustrated schematically in Figure 5.29. The operation of such a mill can be modeled as a perfectly mixed mill with post classification as shown in Figure 5.30.

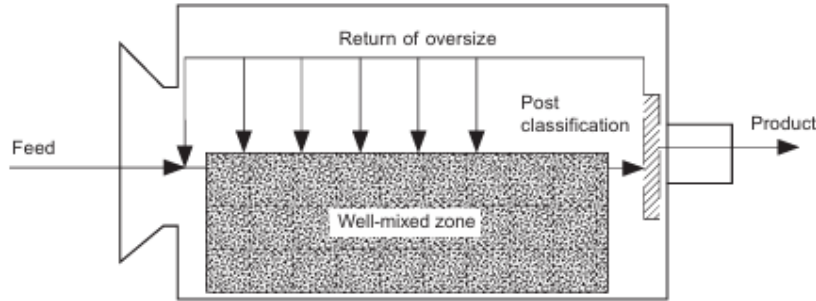


Figure 5.29 The classification mechanism at the mill discharge end recycles larger particles into the body of the mill

Applying Equation 5.84 to the perfectly mixed milling section:

$$m_i = \frac{f_i + \sum_{j=1}^{i-1} b_{ij} k_j \tau' m_j}{1 + \tau' k_i} \tag{5.86}$$

5.12 Models for the specific rate of breakage in ball mills

The utility the kinetic model for breakage depends on the availability of robust models for the specific rate of breakage to describe specific milling conditions. Both functions are strong functions of the milling environment. Factors which affect the rate of breakage are the mill diameter, mill speed, media load and size and particle hold-up.

The most important functional dependence is between the specific rate of breakage and the particle size and methods for the description of this functional dependence are described below. The specific rate of breakage increases steadily with particle size which reflects the decreasing strength of the particles as size increases. This is attributed to the greater density of microflaws in the interior of larger particles and to the greater likelihood that a particular large particle will contain a flaw that will initiate fracture under the prevailing stress conditions in a mill. The decrease in particle strength does not lead to an indefinite increase in the specific rate of breakage. As the particle size becomes significant by comparison to the size of the smallest media particles, the prevailing stress levels in the mill are insufficient to cause fracture and the specific rate of breakage passes through a maximum and decreases with further increase in particle size. Some typical data are shown in Figure 5.35.

5.12.1 The Austin model for the specific rate of breakage

Austin represents the variation of the specific rate of breakage with particle size by the function

$$k(d_p) = \frac{S_1 d_p^\alpha}{1 + (d_p/\mu)^\wedge} \quad (5.127)$$

and it is usual to specify the particle size in mm and the specific rate of breakage in min^{-1} .

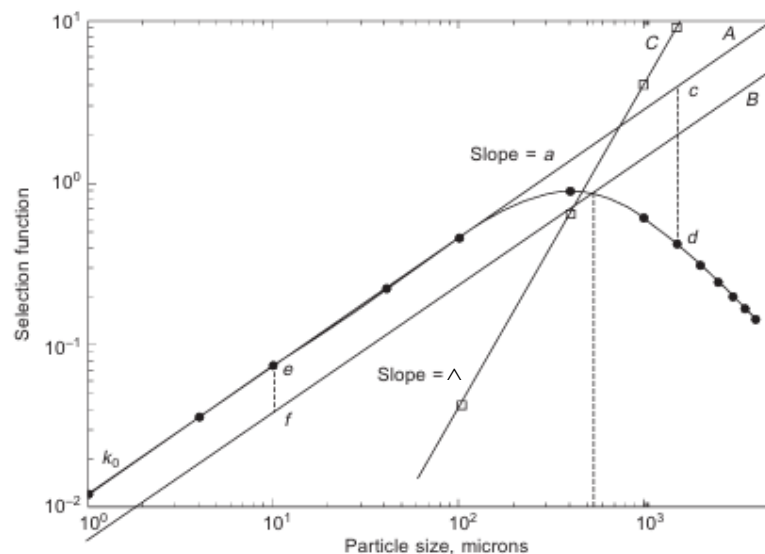


Figure 5.35 Graphical construction to calculate the parameters in Austin's model for the selection function

It is useful to relate the individual parameters in this function to specific features of the graph of $k(d_p)$ plotted against d_p . A typical plot of this function is shown in Figure 5.35 and it has a maximum at a particle size somewhat smaller than the parameter μ which essentially fixes the position of the maximum. The maximum occurs in the specific rate of breakage because as the particles get larger they are less likely to be broken during any typical impact in the mill. The specific particle fracture energy decreases as size increases according to Equation 5.9 but the rate of decrease reduces as the particle size increases and eventually becomes approximately constant for particles larger than a few millimeters for most ores. Consequently the particle fracture energy increases at a rate approximately proportional to the particle mass. Thus for a given impact energy larger particles have a smaller probability of breaking since the energy that a particle absorbs from the impact must exceed its fracture energy otherwise it will not break.

The graphical procedure is implemented as follows:

- (i) Extend the initial straight line portion of the data curve as straight line A having equation

$$k = S_1 d_p \quad (5.128)$$

- (ii) S_1 is evaluated at the intersection of the straight line with the ordinate $d_p = 1$ mm. α is equal to the slope of the line. S_1 is called the selection function at 1 mm.

- (iii) Evaluate ratios such as c/d to form $S_1 d_p / k(d_p)$ at a number of particle sizes as shown.

Plot $(S_1 d_p / k(d_p)) - 1$ against the particle size as shown as line C in Figure 5.35. The slope of the resulting line is equal to Λ because, according to Equation 5.127,

$$\frac{S_1 d_p}{k(d_p)} - 1 = (d_p / \mu)^\Lambda \quad (5.129)$$

Parameter μ can be evaluated in one of two ways.

Construct line B , parallel to line A and passing through a point f which has ordinates equal to $0.5e$.

This line intersects the data curve at abscissa value μ as shown.

Alternatively the size at which $k(d_p)$ is a maximum is given by

$$\frac{dk(d_p)}{dd_p} = 0 \quad (5.130)$$

which implies that

$$\frac{\mu}{d_{p\max}} = \left(\frac{\Lambda}{\alpha} - 1 \right)^{\frac{1}{\Lambda}} \quad (5.131)$$

from which μ can be evaluated when Λ is known.

5.12.3 The Herbst-Fuerstenau model for the specific rate of breakage

The variation of the specific rate of breakage and of the breakage function with milling conditions can be substantially accounted for by describing the variation of these functions with the specific power input into the mill and examining how the specific power input varies with mill size and with design and operating conditions. This indirect method is particularly useful for scaling up the functions from batch scale and pilot scale test work. This is called the energy-specific scale-up procedure and was developed initially by Herbst and Fuerstenau (1980).

The specific rate of breakage out of size class i is proportional to the net specific power input to the mill charge.

$$k_i = S_i^E \frac{P}{M} \quad (5.135)$$

where P is the net power drawn by the mill exclusive of the no-load power that is required to overcome mechanical and frictional losses in the mill. M is the mass of the charge in the mill excluding the media. S_i^E is called the energy-specific selection function for particles in size class i . The energy-specific breakage rate is commonly reported in tonnes/kW hr. The essential feature of the Herbst-Fuerstenau model is that S_i^E is a function of the material only and does not vary with milling conditions nor with mill size. This assumption is equivalent to the postulate that the amount of breakage that occurs inside the mill is proportional to the amount of energy that has been absorbed by the material that is being milled and it is immaterial how the energy is actually imparted to the particles or at what rate. Thus the method can be used to scale up the breakage rates over broad operating ranges. In fact it has been found from experimental observations that S_i^E is a function of the ball media size distribution in the mill. It should be determined in a laboratory experiment using the same ball size distribution as the full-scale mill that is to be modeled or simulated.

It is usually convenient to use Equation 5.135 in the form

$$k_i = S_i^E \frac{P}{W} \quad (5.136)$$

where W is the mass flowrate of solids through the mill. The product k can be used in Equations 5.84, 5.90 and 5.111 and neither the mean residence time of the solids in the mill nor the specific rate of breakage k_i need be known explicitly. This eliminates the need for complex empirical scale-up rules such as Equation 5.132. The ratio P/W , which is measured in kWhr/tonne, is the net specific power consumption in the mill.